Collisions in Weighted Lucas Sums and Differences

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Abstract

We classify collisions among positive weighted Lucas differences and sums with integer coefficient q. For $q \geq 2$, the set $\{L_i - qL_j > 0\}$ has finitely many collisions, and a uniform index bound decides global uniqueness. Infinitely many collisions in $\{qL_i - L_j > 0\}$ and $\{qL_i + L_j\}$ occur for exactly three arithmetic coefficient families: integral Lucas quotients, integral Fibonacci quotients, and half-Lucas coefficients. Otherwise a finite certificate and a signed quotient of two Lucas gaps classify every collision. The sum form has unique representations exactly when both difference forms do. Sign reductions extend the classification to all integer coefficients.

Let $F_0 = 0, F_1 = 1$ and $L_0 = 2, L_1 = 1$, with $u_{n+2} = u_{n+1} + u_n$. Use $F_{-n} = (-1)^{n+1}F_n$ and $L_{-n} = (-1)^n L_n$. Put

$$S_q = \{L_i - qL_j > 0 : i, j \geq 1\}, \quad T_q = \{qL_i - L_j > 0 : i, j \geq 1\}, \quad P_q = \{qL_i + L_j : i, j \geq 1\}.$$

The recurrence gives

$$L_{m+4} - L_m = 5F_{m+2}, \qquad F_a L_b = F_{a+b} + (-1)^b F_{a-b}. \tag{1}$$

Infinite-collision constructions

Theorem 1 (Lucas–Fibonacci quotient coefficients)**.** *Let $m > h \geq 1$ have the same parity, and suppose*

$$q = \begin{cases} L_m/L_h \in \mathbb{Z}, & m, h \text{ odd}, \\ F_m/F_h \in \mathbb{Z}, & m, h \text{ even}. \end{cases}$$

Then $q \geq 2$, and for every $n \geq 1$,

$$qL_{n+m-h} - L_n = qL_{n+m+h} - L_{n+2m} > 0. \tag{2}$$

In particular, T_q and P_q each contain infinitely many repeated positive values.

Proof. For m odd, the Lucas product formula gives

$$L_{n+2m} - L_n = L_m L_{n+m}, \qquad L_{n+m+h} - L_{n+m-h} = L_h L_{n+m}.$$

For m, h even, the Lucas–Fibonacci identity instead gives

$$L_{n+2m} - L_n = 5F_m F_{n+m}, \quad L_{n+m+h} - L_{n+m-h} = 5F_h F_{n+m}.$$

In either case dividing the two identities gives

$$L_{n+2m} - L_n = q(L_{n+m+h} - L_{n+m-h}),$$

which is (2). The right-hand side of the first expression $qL_{n+m-h}-L_n$ is strictly positive and increases with n because $q \geq 2$ and $m-h \geq 1$. The equivalent sum collision is

$$qL_{n+m-h} + L_{n+2m} = qL_{n+m+h} + L_n.$$

□

Theorem 2 (Half-Lucas collision coefficients)**.** *For every integer $r \geq 0$, let $M = 6r+3$ and*

$$q = \frac{L_M}{2} \in \mathbb{Z}.$$

Then T_q and P_q each have infinitely many repeated positive values. More precisely, for every $x \geq M$,

$$qL_x - L_{x-M+1} = qL_{x+3} - L_{x+M+1} > 0, \tag{3}$$

and

$$qL_x + L_{x+M+1} = qL_{x+3} + L_{x-M+1}.$$

The coefficients begin $2, 38, 682, 12238, 219602, \ldots$.

Proof. The Lucas residues modulo 2 have period 3, so L_M is even when $3 \mid M$. As M is odd, the product identity gives

$$L_{x+M+1} - L_{x-M+1} = L_M L_{x+1}.$$

The recurrence also gives $L_{x+3} - L_x = 2L_{x+1}$. Substituting $2q = L_M$ proves (3) and the sum identity. As $x \geq M \geq 3$, $qL_x - L_{x-M+1} > 0$; these values strictly increase with x. □

Norm separation and finite collision bounds

Lemma 3 (Algebraic-norm separation)**.** *For integers $d \geq 1$ and $q \geq 1$, with $\alpha = \varphi$ and $\beta = -\alpha^{-1}$,*

$$|\alpha^d - q| \geq \frac{1}{q+1}. \tag{4}$$

Proof. The conjugate algebraic integers $\alpha^d - q$ and $\beta^d - q$ have nonzero integral product: $\alpha^d = F_d\alpha + F_{d-1}$ is irrational. Since $|\beta^d - q| \leq q+1$, the bound follows. □

For completeness, this estimate also makes every bounded value search finite. If $0 < L_i - qL_j \leq X$, then $i > j$ and Binet's formula gives

$$L_j \leq (q+1)(X+q+1)+1, \qquad L_i \leq X + qL_j.$$

The bounds for $0 < qL_i - L_j \leq X$ follow by exchanging the indices when $j > i$, and by $(q-1)L_i \leq X$ when $j \leq i$ and $q \geq 2$. For sums, $qL_i + L_j \leq X$ directly bounds both indices. Thus for any integer coefficient, all distinct values at most X, their multiplicities (when finite), and their ranks are computed by finite exact arithmetic; negative coefficients reduce by exchanging sum and difference forms.

Proposition 4 (No simultaneous-shift positive collisions)**.** *For every integer $q \geq 2$, there are no indices $i < k$, $j < \ell$ such that*

$$L_{n+i} - qL_{n+j} = L_{n+k} - qL_{n+\ell} > 0 \quad \textit{for every } n \geq 0.$$

Proof. An equality for all shifts implies, by Binet's formula, both its α- and β-coefficient equalities. Put $h = k - i \geq 1$, $e = \ell - j \geq 1$, and $s = i - j$. Positivity for large n forces $\alpha^s > q$, so $s \geq 2$, and

$$\alpha^s(\alpha^h - 1) = q(\alpha^e - 1)$$

forces $e > h$, hence $e \geq 2$. Conjugation gives

$$q|\beta^e - 1| = |\beta|^s|\beta^h - 1| \leq \alpha^{-2}(1 + \alpha^{-1}) = \alpha^{-1}.$$

But $q|\beta^e - 1| \geq 2(1 - \alpha^{-2}) = 2/\alpha > \alpha^{-1}$, a contradiction. □

The global finite-collision bound

A finite threshold is sufficient to compute any term. The stronger statement below provides a bound independent of the value.

Theorem 5 (Uniform complete collision certificate)**.** *Fix an integer $q \geq 2$. Let $t = t(q)$ be the least positive index such that*

$$L_t > 200(q + 1)^4. \tag{5}$$

If two distinct positive representations satisfy

$$L_i - qL_j = L_k - qL_\ell > 0, \qquad i < k,$$

then

$$\boxed{k \leq 6t.} \tag{6}$$

Consequently every S_q with $q \geq 2$ has only finitely many multiply represented positive values. Its complete collision list and global uniqueness can be determined by checking the finitely many pairs $1 \leq j < i \leq 6t$.

Proof. Write $\alpha = \varphi$, $\beta = -\alpha^{-1}$. The equality forces $j < \ell$. Moreover, positivity gives $k > \ell$. Put $d = k - \ell$, $h = k - i$ and $g = k - j$. We prove $k \leq 6t$ by contradiction. The bounds

$$\frac{1}{2}\alpha^n \leq L_n \leq 2\alpha^n \quad (n \geq 0), \qquad L_k - L_i \geq L_{k-2}$$

follow from Binet's formula and monotonicity. Since

$$L_{k-2} \leq L_k - L_i = q(L_\ell - L_j) < qL_\ell,$$

we have $\alpha^d < 4q\alpha^2 < \alpha^t$; thus $1 \leq d < t$. Suppose $k > 6t$. Then $\ell = k - d > 5t$.

Write

$$x = L_k - qL_\ell = \alpha^\ell(\alpha^d - q) + \beta^k - q\beta^\ell.$$

The norm argument in (4) gives $|\alpha^d - q| \geq 1/(q + 1)$. Since $|\beta^k - q\beta^\ell| \leq q + 1$ and $\alpha^\ell > 2(q + 1)^2$, the positivity of x forces $\alpha^d - q > 0$ and

$$x \geq \frac{\alpha^\ell}{2(q + 1)}.$$

But $x < L_i \leq 2\alpha^i$, so $\alpha^{\ell - i} < 4(q + 1) < \alpha^t$, which gives $1 \leq h < 2t$.

Next, the original equality can be rearranged as

$$qL_j = L_i - L_k + qL_\ell = \alpha^k A + \beta^k A',$$

where

$$A = \alpha^{-h} - 1 + q\alpha^{-d}, \qquad A' = \beta^{-h} - 1 + q\beta^{-d}.$$

Both are conjugate algebraic integers, and

$$|A'| \le \alpha^h + 1 + q\alpha^d < 3\alpha^{2t} < \alpha^{3t}.$$

If $A = 0$, then $qL_j = |\beta^k A'| < 1$, an impossibility. Otherwise AA' is a nonzero integer, so $|A| \ge \alpha^{-3t}$. As $k > 6t$ and $qL_j > 0$, this forces $A > 0$ and

$$qL_j > \tfrac{1}{2}\alpha^{k-3t}.$$

Using $qL_j \le 2q\alpha^j$ gives $\alpha^{k-j} < 4q\alpha^{3t} < \alpha^{4t}$, hence $g < 4t$.

Finally, express the collision as

$$0 = L_k - L_i - qL_\ell + qL_j = \alpha^k E + \beta^k E',$$

with the conjugate algebraic integers

$$E = 1 - \alpha^{-h} - q\alpha^{-d} + q\alpha^{-g}, \quad E' = 1 - \beta^{-h} - q\beta^{-d} + q\beta^{-g}.$$

The bounds on d, h, g imply

$$|E'| \le 1 + \alpha^{2t} + q\alpha^t + q\alpha^{4t} < \alpha^{6t}.$$

If $E \ne 0$, the integral norm $EE' \ne 0$ gives $|E| \ge 1/|E'|$. The displayed equality now implies $\alpha^{2k} \le |E'|^2 < \alpha^{12t}$, contrary to $k > 6t$.

If $E = 0$, the equality also forces $E' = 0$. It then holds after every simultaneous translation of all four indices. Because $\alpha^d - q > 0$, the translated differences are positive for all sufficiently large translations, contradicting Proposition 4. Thus neither case is possible, and $k \le 6t$. □

The stationary collision bound

Theorem 6 (Finite stationary pattern criterion)**.** *Fix any integer $q \ge 2$ and choose t as in (5). If*

$$qL_k - L_\ell = qL_i - L_j > 0, \qquad i < k, \tag{7}$$

and $k > 12t$, then all four indices lie within $6t$ of k, apart from a possible upper extension of at most t, and the same equality holds after every *simultaneous nonnegative shift of its indices. Consequently, the set T_q has infinitely many multiply represented positive values if and only if there are indices with*

$$K = 12t + 1, \quad K - 3t \le i < K, \quad K - t \le \ell \le K + t, \quad K - 6t \le j < \ell$$

such that

$$0 < qL_K - L_\ell = qL_i - L_j, \qquad qL_{K+1} - L_{\ell+1} = qL_{i+1} - L_{j+1}. \tag{8}$$

This is a finite, exact, necessary-and-sufficient integer test for the existence of infinitely many positive collisions in T_q. Every collision not belonging to a stationary family has larger first index $k \le 12t$.

Proof. Equation (7) forces $j < \ell$ and $q(L_k - L_i) = L_\ell - L_j$. The bounds $L_{n-2} \le L_n - L_{n-1}$ and $\frac{1}{2}\alpha^n \le L_n \le 2\alpha^n$ yield

$$qL_{k-2} \le L_\ell, \qquad L_{\ell-2} \le qL_k.$$

Consequently, with $\delta = \ell - k$,

$$\alpha^{-\delta} < 4\alpha^2/q < \alpha^t, \qquad \alpha^\delta < 4q\alpha^2 < \alpha^t,$$

so $|\delta| < t$. Assume $k > 12t$. Set

$$x = qL_k - L_\ell = \alpha^k A + \beta^k A', \quad A = q - \alpha^\delta, \quad A' = q - \beta^\delta.$$

The conjugate algebraic integers A, A' have nonzero integral norm, including when $\delta = 0$ (since then $A = q - 1$). As $|A'| < \alpha^{2t}$, we have $|A| > \alpha^{-2t}$. Positivity of x and $k > 12t$ imply $A > 0$ and $x > \frac{1}{2}\alpha^{k-2t}$. Since $x < qL_i \le 2q\alpha^i$, the gap $h = k - i$ satisfies $0 < h < 3t$.

Next,

$$L_j = qL_i - qL_k + L_\ell = \alpha^k B + \beta^k B',$$

where $B = q\alpha^{-h} - q + \alpha^\delta$ and $B' = q\beta^{-h} - q + \beta^\delta$. The conjugates are algebraic integers and $|B'| < \alpha^{5t}$. If $B = 0$, then $L_j < 1$, a contradiction; otherwise their integral norm gives $|B| > \alpha^{-5t}$. As $k > 12t$ and $L_j > 0$, we obtain $L_j > \frac{1}{2}\alpha^{k-5t}$. Thus $g = k - j < 6t$. Since $j < \ell$, also $g > -t$.

Now the collision is

$$0 = qL_k - qL_i - L_\ell + L_j = \alpha^k E + \beta^k E',$$

where

$$E = q - q\alpha^{-h} - \alpha^\delta + \alpha^{-g}, \quad E' = q - q\beta^{-h} - \beta^\delta + \beta^{-g}.$$

Both are algebraic integers and $|E'| < \alpha^{8t}$. If $E \neq 0$, the integral norm gives $|E| > 1/|E'|$ and the equality forces $\alpha^k \le |E'| < \alpha^{8t}$, contradicting $k > 12t$. Therefore $E = E' = 0$, and the identity holds under all simultaneous shifts. The positive leading coefficient A makes its common values positive and unbounded under increasing shifts.

Every infinite sequence of distinct collision values must have unbounded k, and so contains a stationary identity of the preceding form. Translate its indices to $k = K = 12t + 1$. The established gap bounds give precisely the finite ranges in (8). The leading coefficient $A > 0$ ensures positivity at K. Conversely, the two equalities in (8) give consecutive zeros of a Lucas-recursive difference. They therefore give the identity at all shifts, and its positive values become arbitrarily large. This proves the criterion and the asserted bound for nonstationary collisions. □

Proposition 7 (Finite certificates for all three forms)**.** *Fix $q \ge 2$ and t as in (5). If T_q has no stationary collision, every positive collision of T_q or P_q has all indices at most $13t$. If a stationary collision exists, both forms have a positive collision with all indices at most $13t + 1$. In either case global uniqueness is decided by a finite search. For S_q, the bound is $6t$. For $q = 1$ all three forms have infinitely many collisions; for $q = 0$, T_0 is empty and S_0, P_0 have infinite representation multiplicities. For $q = -m < 0$, T_q is empty, $S_q = P_m$, and $P_q = S_m$ after exchanging index roles.*

Proof. Theorem 5 settles S_q. When T_q has no stationary collision, Theorem 6 puts its larger first index at most $12t$. Positivity gives $L_j < qL_i$, and hence $j < i + t$, so all indices are at most $13t$. A sum collision with unequal first indices has a corresponding signed difference

$$v = qL_k - L_j = qL_i - L_\ell.$$

If $v > 0$, this is a collision in T_q; if $v < 0$, its negative is one in S_q. The case $v = 0$ would give two different representations of the same Lucas ratio $q > 1$, contrary to Lemma 10. Thus the same index bound holds for P_q. If T_q has a stationary collision, Theorem 6 supplies one at $k = 12t + 1$ with all other indices at most $13t + 1$; swapping subtracted indices gives the required sum collision. The remaining signs follow from the recurrence and the definitions. □

Explicit classification of infinite collisions

The quotient and half-Lucas identities exhaust the infinite-collision coefficients. We classify the index gaps of every stationary identity.

Lemma 8 (Rigidity of the two index gaps)**.** *Suppose $q \geq 2$, $i < k$, $j < \ell$, and*

$$q(L_{n+k} - L_{n+i}) = L_{n+\ell} - L_{n+j} \qquad (n \geq 0). \tag{9}$$

Set $h = k - i$, $e = \ell - j$, and $d = i - j$. The only possibilities are:

1. *$h = 2r$, $e = 2m$, $m > r \geq 1$, with $m \equiv r \pmod 2$, $d = m - r$, and either $q = L_m/L_r$ for m, r odd or $q = F_m/F_r$ for m, r even;*
2. *$h = 1$, $e = 2m$, $m \geq 3$ odd, $d = m + 1$, and $q = L_m$;*
3. *$h = 3$, $e = 2m$, $m \geq 3$ odd and divisible by 3, $d = m - 1$, and $q = L_m/2$;*
4. *$(h, e, d, q) = (1, 3, 2, 2)$ or $(2, 3, 0, 2)$.*

Every identity on this list is realized, with positive values and arbitrarily large simultaneous shifts.

Proof. Let $\alpha = \varphi$, $\beta = -\alpha^{-1}$, and set

$$C(s) = L_s - 1 - (-1)^s, \qquad \Phi(s) = \frac{1 + \alpha^{-s}}{1 - \alpha^{-s}} \quad (s \geq 1).$$

The two conjugate Binet coefficient equations of (9) give

$$q\alpha^i(\alpha^h - 1) = \alpha^j(\alpha^e - 1), \qquad q\beta^i(\beta^h - 1) = \beta^j(\beta^e - 1).$$

Their norms yield

$$q^2 C(h) = C(e), \qquad i \equiv j \pmod 2. \tag{10}$$

In particular $e > h$, and $d = i - j$ is even. Dividing the two Binet equations, and using $\beta = -\alpha^{-1}$, gives the exact relation

$$\alpha^{2d} = \alpha^{e-h} \frac{\Phi(h)^{[h \text{ odd}]}}{\Phi(e)^{[e \text{ odd}]}}. \tag{11}$$

The function $\Phi(s)$ decreases strictly to 1; direct calculation gives

$$\Phi(1) = \alpha^3, \qquad \Phi(3) = \alpha. \tag{12}$$

If h, e are both even, (11) gives $2d = e - h$. Write $h = 2r$, $e = 2m$. Since d is even, $m \equiv r \pmod 2$. For even indices,

$$C(2s) = L_{2s} - 2 = \begin{cases} L_s^2, & s \text{ odd}, \\ 5F_s^2, & s \text{ even}. \end{cases}$$

Equation (10) gives the two quotient conditions in case 1.

If both gaps are odd, put $u = (e - h)/2$. Equation (11) becomes

$$\alpha^{2(d-u)} = \Phi(h)/\Phi(e).$$

When $h \geq 3$ this ratio lies strictly between 1 and $\Phi(3) = \alpha < \alpha^2$, which is impossible for an integral exponent $2(d - u)$. When $h = 1$, the ratio is at least $\Phi(1)/\Phi(3) = \alpha^2$, with equality at $e = 3$; for

$e > 3$ it lies strictly between α^2 and $\alpha^3 < \alpha^4$. Therefore $h = 1, e = 3$; then $d = 2$ and (10) gives $q = 2$.

If h is odd and e is even, the extra factor in (11) is $\Phi(h)$. For odd $h \geq 5$, the right side lies strictly between consecutive powers α^{e-h} and α^{e-h+1}, of which only the second has an even exponent. This is impossible. For $h = 1$, (12) gives $2d = e + 2$; for $h = 3$ it gives $2d = e - 2$. Putting $e = 2m$ in the norm equation then gives, respectively, $q = L_m$ or $q = L_m/2$ when m is odd. Indeed m even would require an irrational positive multiple of $\sqrt{5}$ to be the integer q. In the second case, integrality of $L_m/2$ is equivalent to $3 \mid m$; the condition $q \geq 2$ also excludes $m = 1$.

Finally, for even h, odd e, (11) has the factor $1/\Phi(e)$. For $e \geq 5$, the exponent lies strictly between the consecutive integers $e - h - 1$ and $e - h$; the former is even, and no even exponent is possible. Thus $e = 3$, necessarily $h = 2$, and (12) gives $d = 0$ and $q = 2$.

For sufficiency, case 1 is the quotient identity (2). Cases 2 and 3 follow from $L_{n+2m} - L_n = L_m L_{n+m}$ for odd m, together with $L_{n+1} - L_n = L_{n-1}$ and $L_{n+3} - L_n = 2L_{n+1}$. The last two cases are the corresponding $q = 2$ identities. All hold for every sufficiently large common shift, with strictly increasing positive values. □

Theorem 9 (Explicit infinite-collision coefficient classification)**.** *For every integer $q \geq 2$, the set T_q has infinitely many multiply represented positive values if and only if q belongs to the following union:*

$$\begin{aligned} &\{L_m/L_r \in \mathbb{Z} : m > r \geq 1,\ m, r \text{ odd}\} \\ \cup\ &\{F_m/F_r \in \mathbb{Z} : m > r \geq 2,\ m, r \text{ even}\} \\ \cup\ &\{L_{6r+3}/2 : r \geq 0\}. \end{aligned} \tag{13}$$

The same equivalence holds for P_q. Every other integer $q \geq 2$ has only finitely many collisions in T_q and P_q.

Proof. Each coefficient in (13) has an infinite family of positive collision values by Theorems 1 and 2. Conversely, if T_q has infinitely many collision values, Theorem 6 forces a collision holding under every common index shift. Lemma 8 puts its coefficient in one of the three families: cases 2 and 4 are subsumed by the first or third family, and case 3 is the third family. The equivalence for P_q and finiteness outside these families follow from Proposition 7. □

Finite-collision coefficient spectra

The preceding global bounds can also classify the coefficients with finitely many collisions without computing any ordered set of positive values. The sole arithmetic objects needed are differences of two Lucas numbers and the sign of a two-by-two determinant.

For integers $1 \leq a < b$ put $D(a, b) = L_b - L_a > 0$. Given two index pairs $a < b$ and $c < d$, define

$$X = D(a, b), \quad Y = D(c, d), \qquad \Delta(a, b; c, d) = L_a L_d - L_b L_c.$$

The integrality condition $Y \mid X$ generates the candidate coefficient $q = X/Y$.

Lemma 10 (Uniqueness of positive Lucas ratios)**.** *For $a > b \geq 1$, $c > d \geq 1$, the equality $L_a/L_b = L_c/L_d$ implies $(a, b) = (c, d)$.*

Proof. Set $R_h(n) = L_{n+h}/L_n$, $\alpha = \varphi$. Binet's formula gives, for $n \geq 2, h \geq 1$,

$$\frac{4}{3\alpha}\alpha^h \leq R_h(n) \leq \frac{2}{\alpha}\alpha^h, \tag{14}$$

since

$$\frac{1-\alpha^{-6}}{1+\alpha^{-4}} = \frac{4}{3\alpha}, \qquad \frac{1+\alpha^{-6}}{1-\alpha^{-4}} = \frac{2}{\alpha}.$$

The intervals for distinct h are disjoint, because $4/3 > 2/\alpha$. For fixed h,

$$L_m L_{n+h} - L_n L_{m+h} = 5(-1)^{n+1} F_{m-n} F_h \neq 0 \qquad (m > n \geq 2).$$

Thus all ratios with $n \geq 2$ are distinct.

For $n = 1$, write $R_h(1) = L_s$, $s = h + 1 \geq 2$. Since

$$(1-\alpha^{-4})\alpha^s \leq L_s \leq (1+\alpha^{-4})\alpha^s,$$

the inequalities $\alpha(1-\alpha^{-4}) > 2/\alpha$ and $1+\alpha^{-4} < 4/3$ show that $L_s = R_e(m)$, $m \geq 2$, would force $e = s$. But then the Lucas product formula gives

$$L_s L_m - L_{m+s} = (-1)^s L_{m-s} \neq 0,$$

including $m < s$ via the standard extension to negative Lucas indices. Ratios with both denominators L_1 are strictly increasing with their numerators. □

Theorem 11 (Signed-gap coefficient spectra)**.** *Fix $q \geq 2$. The following statements are exact necessary-and-sufficient conditions for the existence of a multiply represented* positive *value:*

1. *S_q has a collision if and only if some $1 \leq a < b$, $1 \leq c < d$ satisfy $D(a,b) = qD(c,d)$ and $\Delta(a,b;c,d) > 0$.*
2. *T_q has a collision if and only if such indices satisfy $\Delta(a,b;c,d) < 0$.*
3. *P_q has a collision if and only if such indices exist, regardless of the sign of Δ.*

All three tests can be restricted to the common finite index window

$$\boxed{1 \leq a, b, c, d \leq 13t + 1,} \tag{15}$$

where $t = t(q)$ is defined by (5). The two strict determinant signs construct all collision witnesses using exact integer arithmetic.

Proof. If $D(a,b) = qD(c,d)$, then the two unequal pairs of indices give the sum collision

$$qL_c + L_b = qL_d + L_a > 0.$$

Furthermore,

$$L_a - qL_c = L_b - qL_d = \frac{\Delta}{D(c,d)},$$

and

$$qL_c - L_a = qL_d - L_b = -\frac{\Delta}{D(c,d)}.$$

This proves that positive, negative, and zero signs have precisely the asserted consequences. Conversely, any collision in S_q has first indices $a < b$ and second indices $c < d$; its equality and positivity give the divisor equation and $\Delta > 0$. For a T_q collision, label the first indices $c < d$ and second indices $a < b$; the same operations yield $\Delta < 0$. For P_q, the two first and second indices necessarily run in opposite directions; relabeling them gives the divisor equation. Thus all three tests are equivalent, not merely sufficient.

If $\Delta = 0$, the divisor equation implies $L_a = qL_c$ and $L_b = qL_d$. These are two distinct representations of the same Lucas ratio $q \geq 2$, contrary to Lemma 10. Thus no zero-determinant coefficient candidate exists.

For the common index bound, Theorem 5 places all S_q collision indices at most $6t$. If q has no stationary T_q collision, Proposition 7 places every T_q and P_q collision inside $13t$. Otherwise Theorem 6 supplies a positive stationary T_q collision at $k = 12t+1$, whose other three indices belong to $[1, 13t+1]$; exchanging subtracted indices also supplies a P_q collision inside that window. In all cases, if a collision exists, a certificate satisfying (15) exists. □

Define two explicitly parametrized sets of positive integer coefficients:

$$\boxed{\mathcal{C}_\pm = \left\{ \frac{D(a,b)}{D(c,d)} \in \mathbb{Z}_{\geq 2} : a < b,\ c < d,\ \pm\Delta(a,b;c,d) > 0 \right\}.} \tag{16}$$

All four indices are positive. Thus S_q has a positive collision precisely for $q \in \mathcal{C}_+$; T_q precisely for $q \in \mathcal{C}_-$; and P_q precisely for $q \in \mathcal{C}_+ \cup \mathcal{C}_-$. Let $\mathcal{I}$ be the three explicit coefficient families in (13). By Theorem 9, the *finitely collided but nonunique* coefficients are

$$\begin{array}{c|l} S & \mathcal{C}_+ \\ T & \mathcal{C}_- \setminus \mathcal{I} \\ P & (\mathcal{C}_+ \cup \mathcal{C}_-) \setminus \mathcal{I}. \end{array}$$

The complements give global uniqueness, while $\mathcal{I}$ gives infinitely many collision values for T and P. For S, all collisions are finite at every $q \geq 2$.

Corollary 12 (Sum uniqueness from both differences)**.** *For every integer $q \geq 2$, all positive values in P_q have unique index-pair representations if and only if this holds simultaneously in S_q and T_q.*

Proof. Theorem 11 partitions all positive P_q collision witnesses into the two strict signs, respectively equivalent to S_q and T_q collisions. The zero sign is impossible by Lemma 10. □

Proposition 13 (Equal-gap parity rule)**.** *For $h \geq 1$ and $1 \leq c < a$, suppose*

$$q = \frac{L_{a+h} - L_a}{L_{c+h} - L_c} \in \mathbb{Z}_{\geq 2}.$$

Then P_q has repeated positive values. If c is odd, S_q has a positive collision; if c is even, T_q has a positive collision. In particular, each of the infinite families

$$\boxed{q = \frac{L_{3r}}{2}\ (r \geq 1), \qquad q = \frac{L_{4r+2}}{3}\ (r \geq 1), \qquad q = \frac{F_{3r}}{2}\ (r \geq 2)}$$

consists entirely of coefficients for which S_q has a finite nonempty collision set.

Proof. The Lucas–Fibonacci product identity gives

$$\boxed{L_a L_{c+h} - L_c L_{a+h} = 5(-1)^{c+1} F_{a-c} F_h.} \tag{17}$$

For $a > c$ and $h \geq 1$, both Fibonacci factors are positive. Theorem 11 therefore proves the parity rule.

For the three families choose $c = 1$. Using the elementary identities $L_{n+1} - L_n = L_{n-1}$, $L_{n+2} - L_n = L_{n+1}$, and $L_{n+4} - L_n = 5F_{n+2}$, take

$$(h, a) = (1, 3r+1), \qquad (2, 4r+1), \qquad (4, 3r-2),$$

respectively (the last has $r \geq 2$). The denominators $D(1, 1+h)$ are $2, 3, 10$. Divisibility follows from the period-three Lucas/Fibonacci parities and the Lucas residues $L_{4r+2} \equiv 0 \pmod 3$. Theorem 5 proves that every fixed S_q has only finitely many collisions. □

A family with globally unique differences

Theorem 14. *Fix $r \geq 1$ and set $q = 5F_{2r}$. Then*

$$L_i - qL_j > 0 \quad \Longleftrightarrow \quad (i,j) = (2r+2,1) \quad \text{or} \quad i \geq 2r+3,\ 1 \leq j \leq i-2r-2.$$

Every positive integer in S_q has precisely one representation.

Proof. The recurrence yields

$$q = 5F_{2r} = L_{2r+2} - L_{2r-2}, \quad q - L_{2r+1} = L_{2r-1} > 0, \quad L_{2r+2} - q = L_{2r-2} > 0. \tag{18}$$

Thus $j = 1$ first becomes admissible at $i = 2r+2$. For $j \geq 2$ put $A_j = L_{j+2r+2} - qL_j$ and $B_j = qL_j - L_{j+2r+1}$. Both obey the Lucas recurrence in j. Checking two initial values in r and using the recurrence in steps of two gives

$$A_2 = L_{2r+4} - 3q = L_{2r-4} > 0, \quad A_3 = A_2 + A_1 > 0, \quad A_1 = L_{2r+3} - q > 0,$$

with $L_{-2} = 3$ when $r = 1$. For B_j, the first two terms of the sequences $B_2(r)$ and $B_3(r)$ are

$$(B_2(1), B_2(2)) = (4, 16), \qquad (B_3(1), B_3(2)) = (2, 13).$$

Each satisfies $b_{r+2} = 3b_{r+1} - b_r$ as a function of r. An increasing positive initial pair remains increasing and positive. Therefore $B_2, B_3 > 0$ for all $r \geq 1$; the recurrence in j then gives $A_j, B_j > 0$ for all $j \geq 2$. Hence $L_{j+2r+1} < qL_j < L_{j+2r+2}$ for $j \geq 2$, establishing the stated index region.

Let $\mathcal{B}_i$ be the positive values with first index i. They are distinct within each block, and $\max \mathcal{B}_i = L_i - q$ for $i \geq 2r+2$. Using (1) with $b = i - 2r + 2$, the next block of the same residue class satisfies

$$\begin{aligned} \min \mathcal{B}_{i+4} - \max \mathcal{B}_i &= L_{i+4} - L_i - qL_{i-2r+2} + q \\ &= q + 5F_{i-4r+2} > 0. \end{aligned} \tag{19}$$

For a nonnegative Fibonacci index the inequality is immediate. For a negative index, $i \geq 2r+2$ implies $0 < |i - 4r + 2| \leq 2r - 4 < 2r$, so $|F_{i-4r+2}| < F_{2r}$. Moreover $A_{j+4} > A_j$ follows from the strictly positive Lucas recurrence; thus blocks separated by any positive multiple of four are strictly ordered.

Finally, $(L_1, L_2, L_3, L_4) \bmod 5 = (1, 3, 4, 2)$ is the repeating residue cycle. Since $5 \mid q$, two equal differences must have first indices congruent modulo four. They cannot belong to different such blocks, by (19), and the same block has distinct values. □